Tarek Ibrahim

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Education

Carleton University — Ottawa, Ontario
Bachelor of Engineering, Software Engineering (Co-op) — 4th Year — ***Graduation: April 2027***
Relevant Courses: Algorithms and Data Structures, Computer Architecture, Software Design, Operating Systems, Software Requirements, Advanced C++ Programming, Computer Networks, Real-Time Embedded Systems, Database Management

Skills

Languages:	C++, C, Python, Go, TypeScript, HTML/CSS, Java, SQL, Bash
Libraries:	CUDA/HIP, OneAPI, LLVM, Boost, GTest, OpenCV, JAX, PyTorch, Gym, React, Next.js, Gin, gRPC
Development Tools:	CMake, Valgrind, Wireshark, Docker, Conda, Kafkacat, SLURM, Gitlab, GCloud, Atlassian Suite

Relevant Experience

Software Developer — May 2025 – December 2025
Searidge Technologies (SRT) — *Ottawa, ON / Hybrid*
- Developed and maintained safety-critical C++ microservices for air traffic control systems, ensuring high reliability and performance in real-time environments while conforming to ED-109A software integrity assurance standards.
- Rewrote and modularized a multithreaded logger using idiomatic modern C++ practices to improve throughput, maintainability, and concurrency safety for airport software systems deployed in 21 countries and 56 sites across the globe.
- Built a multithreaded video stabilization microservice using Kafka, OpenCV, and CUDA runtime for real-time streaming.
- Drove technical improvements by contributing to code reviews through GitLab and researching new frameworks.

Software Developer — January 2025 – April 2025
Agriculture and Agri-Food Canada (AAFC) — *Ottawa, ON / Hybrid*
- Deployed Nextflow bioinformatics pipelines (RNAseq, Ampliseq, Sarek) on the 2 most powerful supercomputers in Canada.
- Engineered an in-house solution to enable flexible LLM inference by deploying Ollama via Apptainer on 8x NVIDIA A100 Tensor Core GPUs, overcoming previous limitations, and serving researchers across multiple Canadian departments.
- Improved documentation and technical guides for HPC workflows, including CUDA, SLURM, TBB, and OpenMP.
- Provided support by answering technical questions and resolving issues for users of the supercomputer infrastructure.

Software Developer — May 2024 – August 2024
James Evan & Associates (JEA) — *Victoria, BC / Remote*
- Eliminated legacy enzyme-adapter bottleneck by migrating 500+ tests to React Testing Library enabling critical updates.
- Upgraded core Typescript dependencies and runtime environments, resolving breaking changes, deprecated features, API modifications to ensure integration with CI/CD pipelines in Jenkins.

Undergraduate ML Researcher — May 2024 – May 2025
Carleton University — *Ottawa, ON*
- Co-authored "LookWhere? Efficient Visual Recognition by Learning Where to Look and What to See from Self-Supervision," **accepted** for publication at NeurIPS 2025, a top-tier conference in machine learning and AI. Preprint available on arXiv.
- Enabled large-scale Vision Transformer (ViT) training by deploying JAX & PyTorch codebases on Google Cloud TPU pods

Teaching Assistant — September 2024 – Present
Carleton University — *Ottawa, ON*
- **Digital Systems**: Mentored and graded 90+ students in weekly labs; taught circuit design (flip-flops, ALU's, and counters)
- **Foundations of Programming**: Taught C programming, memory management, pointers, and abstract data structures.
- **Computer Architecture**: Assessed students on assembly, finite state machines, databus management, and peripheral I/O.

Leadership Experience

Vice President of Technology & Projects — September 2024 – September 2025
Carleton Artificial Intelligence Society (CUAIS) — *Ottawa, ON*
- Founded the club and Led a team of 15+ in collaborative open source projects growing the club to 50+ active members.

Projects

CuMind: Reinforcement Learning Library | *Python, JAX, Flax NNX, Gymnasium, Google Cloud TPU*
- Built a modular JAX based RL library integrating Gymnasium environments for training on Google Cloud Platform (GCP).
- Added configurable profiling, tracing, logging, checkpointing, and PRNG controls to support CPU/GPU/TPU backends.
- Reimplemented Google DeepMind's MuZero algorithm by combining representation, dynamics, and prediction networks with Monte Carlo Tree Search (MCTS) for model-based planning across diverse domains (discrete and continuous).

Vibecheck: Tweet Guessing Game | *Go, Redis, PostgreSQL, Gin, Docker, React*
- Developed a full-stack web app with CRUD operations on 30k twitter tweets and extended the RESTful API to include a party guessing game for users to guess the *vibe* of a tweet, accessible via a frontend.